Aaryaman Kalani

B.E. Computer Science & MSc Physics, BITS Pilani • Expected June 2027
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Research Interests

AI Safety & Alignment • Model Interventions • Machine Unlearning
Mechanistic Interpretability • LLM Post-Training

I am interested in uncovering what model interventions, such as unlearning, continual learning and post-training, really change inside a model, and when behavioral evidence misleads. I want to study how interpretability can help us understand these interventions, and vice-versa.

Publications & Preprints

The BatchNorm Illusion: Diagnosing Normalization Artifacts in Machine Unlearning Evaluation

Aaryaman Kalani, Murari Mandal, Dhruv Kumar, Mohan Kankanhalli, Yash Sinha
Advances in Neural Information Processing Systems (NeurIPS) 2026, Main Track **Accepted**

- Diagnosed normalization artifacts across nine published machine-unlearning methods.
- Reversed up to 78 percentage points of reported forgetting with one weight-preserving pass over retain data.
- Proved the pass is a fixed-point operator on weights, separating normalization-driven measurement bias from residual leakage.
- Recovered most hidden accuracy using only 10 unlabeled images in an attacker setting.
- A GroupNorm control eliminated the effect (0.00 pp), ruling out architecture, dataset, and hyperparameter confounds.

Forgotten Compared to What? A Calibrated Audit of Class Unlearning

Aaryaman Kalani, Murari Mandal, Dhruv Kumar, Mohan Kankanhalli, Yash Sinha
International Conference on Learning Representations (ICLR) 2027 **Under review**

- Introduced woken recall: one logit offset, set on retained data at a 5% false-positive rate, compared with utility-matched retain-only models.
- Proved the audit is invariant to class-bias shifts, so suppressing the removed logit cannot hide recovery.
- Across 480 CIFAR-10 checkpoints from 16 methods, 347 of 372 supported ones recovered more of the removed class than matched references (median +49.5 points).

Style, Not Strategy: Discriminating Post-Training Displacement from Its Confounds in LLM Price Negotiation

Aaryaman Kalani, Dhruv Kumar, Yash Sinha
Transactions on Machine Learning Research (TMLR), 2026 **Accepted**

- Isolated a format-controlled change in negotiation behavior after LLM post-training.
- Designed three controls: format wedge, teacher-forced firing, and strategy-vs-style tests.
- A best-response audit found no evidence of improved opponent modeling, pointing instead to stylistic adaptation.

RASE: Retain-Agnostic Machine Unlearning via Activation Subspace Projection

Aaryaman Kalani, Dhruv Kumar, Mohan Kankanhalli, Murari Mandal, Yash Sinha
Machine Unlearning for Vision (MUV) Workshop, CVPR 2026 **Non-archival**

- Developed retain-agnostic unlearning using a one-time offline SVD of retain activations to constrain forget gradients to the retain-orthogonal complement.
- Reached 92.0% mean retain accuracy vs. 41.7% for gradient ascent across six CIFAR-10 forget classes.

Education

Birla Institute of Technology and Science (BITS) Pilani Pilani Campus
MSc Physics & B.E. Computer Science (dual degree) Oct 2022–Jun 2027
CGPA: 7.59/10
Selected coursework: Probability & Statistics, Applied Stochastic Processes, Linear Algebra, Deep Learning, Introduction to LLMs, Foundations of Data Science.

Selected Research Projects

Stress-testing Representation Readouts with the Jacobian Lens
Self-directed; mechanistic interpretability; Qwen3.5-9B Aug–Sep 2026

- Tested whether readout drift tracks prompt-content removal; results instead supported distance from the activation manifold.
- Scored six ablation conditions in lens output space with JS divergence, top-10 Jaccard, and median rank; no learned evaluator.
- A no-removal control drifted less than content removal at all three tested layers.
- Temperature calibration and a $J = I$ twin ruled out sharpness and lens-specific explanations.

DiplomacyArena: LLM Multi-agent Negotiation with Belief Tracking
Advisor: Dr. Dhruv Kumar Feb–Apr 2025

- Built a LangGraph environment with episodic RAG and Theory-of-Mind belief tracking.
- Raised ultimatum-game success to 100% without retraining.

Quantization Effects on Reasoning in Small Language Models
NF4 quantization; QLoRA; GRPO Jan–Apr 2026

- Measured NF4 degradation on DeepSeek-R1-Distill-Qwen-1.5B across two reasoning benchmarks.
- Restored GSM8K accuracy to 80% using QLoRA adapters, with greedy-decoding controls.

Research Experience

Research Intern, IIT Bombay
Advisor: Dr. Shankaranarayanan, Institute Chair Professor May–Oct 2024

- Built a Python framework for the circular restricted three-body problem; analyzed Lagrange-point stability via linearization and eigenvalue methods.

Technical & Research Skills

Programming: Python, C++, C, Java, SQL, MATLAB, Mathematica.
ML / Scientific Computing: PyTorch, NumPy, SciPy, Pandas, Scikit-Learn, QLoRA, quantization, numerical simulation.
Research: Machine unlearning, mechanistic interpretability, LLM post-training evaluation, multi-agent RL.

Service & Leadership

Signatory, Corroboration and Review Committee
BITS Pilani Oct 2023–Apr 2025

- Selected as one of 3 students from 1100+ to oversee campus-union finances.
- Authorized INR 10 Cr+ in transactions and helped reverse an INR 4.1 Cr deficit through fiscal restructuring.